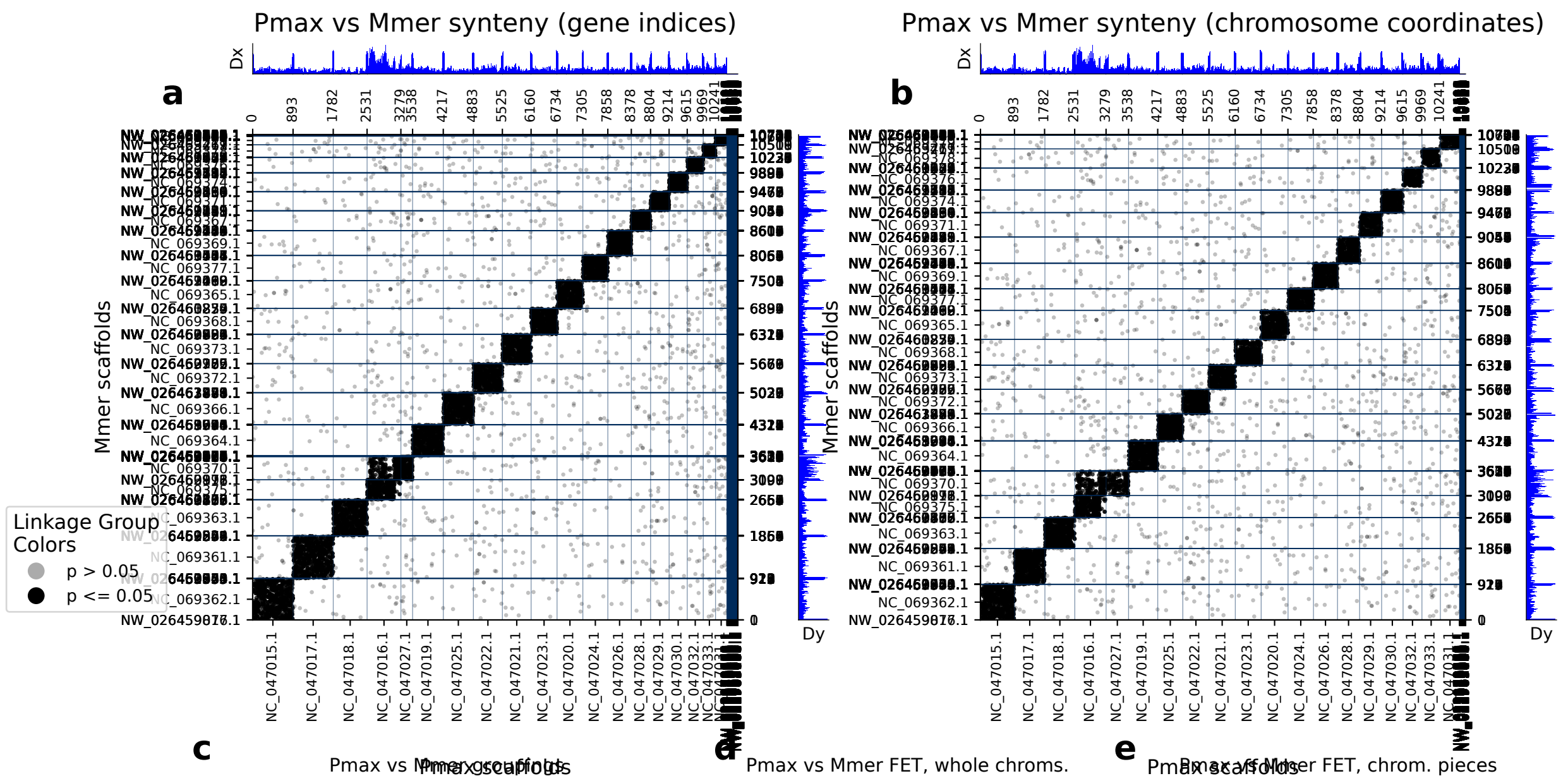


diamond: 10737 orthologs on 237 Pmax scaffolds and 214 Mmer scaffolds.



	Pmax scaf	Mmer scaf	p (FET)	Count
0	NC_047017.1	NC_069361.1	0.00e+00	801
1	NC_047015.1	NC_069362.1	0.00e+00	788
2	NC_047018.1	NC_069363.1	0.00e+00	689
3	NC_047025.1	NC_069366.1	0.00e+00	585
4	NC_047019.1	NC_069364.1	0.00e+00	583
5	NC_047021.1	NC_069373.1	0.00e+00	542
6	NC_047022.1	NC_069372.1	0.00e+00	525
7	NC_047020.1	NC_069365.1	0.00e+00	496
8	NC_047023.1	NC_069368.1	0.00e+00	492
9	NC_047024.1	NC_069377.1	0.00e+00	477
10	NC_047026.1	NC_069369.1	0.00e+00	452
11	NC_047016.1	NC_069375.1	0.00e+00	368
12	NC_047028.1	NC_069367.1	0.00e+00	342
13	NC_047030.1	NC_069374.1	0.00e+00	334
14	NC_047029.1	NC_069371.1	0.00e+00	330
15	NC_047032.1	NC_069376.1	0.00e+00	257
16	NC_047016.1	NC_069370.1	3.11e-147	241
17	NC_047033.1	NC_069378.1	1.63e-283	196
18	NC_047027.1	NC_069370.1	1.74e-175	176
19	NC_047031.1	NC_069379.1	2.08e-193	137
20	NW_022980724.1	NC_048487.1	2.46e-07	3
21	NW_022979575.1	NW_026462340.1	7.92e-03	2

C	Pmax scaf	Mmer scaf	p (FET)	Count
0	NC_047017.1 : 0-end	NC_069361.1 : 0-end	0.00e+00	801
1	NC_047015.1 : 0-end	NC_069362.1 : 0-end	0.00e+00	788
2	NC_047018.1 : 0-end	NC_069363.1 : 0-end	0.00e+00	689
3	NC_047025.1 : 0-end	NC_069366.1 : 0-end	0.00e+00	585
4	NC_047019.1 : 0-end	NC_069364.1 : 0-end	0.00e+00	583
5	NC_047021.1 : 0-end	NC_069373.1 : 0-end	0.00e+00	542
6	NC_047022.1 : 0-end	NC_069372.1 : 0-end	0.00e+00	525
7	NC_047020.1 : 0-end	NC_069365.1 : 0-end	0.00e+00	496
8	NC_047023.1 : 0-end	NC_069368.1 : 0-end	0.00e+00	492
9	NC_047024.1 : 0-end	NC_069377.1 : 0-end	0.00e+00	477
10	NC_047026.1 : 0-end	NC_069369.1 : 0-end	0.00e+00	452
11	NC_047016.1 : 0-end	NC_069375.1 : 0-end	0.00e+00	368
12	NC_047028.1 : 0-end	NC_069367.1 : 0-end	0.00e+00	342
13	NC_047030.1 : 0-end	NC_069374.1 : 0-end	0.00e+00	334
14	NC_047029.1 : 0-end	NC_069371.1 : 0-end	0.00e+00	330
15	NC_047032.1 : 0-end	NC_069376.1 : 0-end	0.00e+00	257
16	NC_047016.1 : 0-end	NC_069370.1 : 0-end	3.11e-14	721
17	NC_047033.1 : 0-end	NC_069378.1 : 0-end	1.63e-28	3196
18	NC_047027.1 : 0-end	NC_069370.1 : 0-end	1.74e-17	5176
19	NC_047031.1 : 0-end	NC_069379.1 : 0-end	2.08e-19	1317
20	NW_022980724.1 : 0-end	NC_048487.1 : 0-end	2.46e-07	3
21	NW_022979575.1 : 0-end	NW_026462340.1 : 0-end	7.92e-03	2

(a) depicts the chromosomes/scaffolds of Pmax (x-axis) plotted against the chromosomes/scaffolds of Mmer (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 10737 orthologs found between 237 Pmax scaffolds and 214 Mmer scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. *Nature Ecology & Evolution* 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chromosome FET p-value, and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.